Irfan Hussain (230123081)
Mark: 9/15

+74/1/34+

MA 324
STATISTICAL INFERENCE AND MULTIVARIATE ANALYSIS
IIT GUWAHATI

QUIZ III
16:05–16:50 IST
APRIL 08, 2026

*Answers must be given **exclusively on this sheet**: answers given on the other sheets **will be ignored**. This question paper has **6 questions** and 2 pages.*

NAME AND ROLL NUMBER

Please mark your Roll No:

○0	○0	●0	○0	○0	○0	●0	○0	○0
○1	○1	○1	●1	○1	○1	○1	○1	●1
●2	○2	○2	○2	●2	○2	○2	○2	○2
○3	●3	○3	○3	○3	●3	○3	○3	○3
○4	○4	○4	○4	○4	○4	○4	○4	○4
○5	○5	○5	○5	○5	○5	○5	○5	○5
○6	○6	○6	○6	○6	●6	○6	○6	○6
○7	○7	○7	○7	○7	○7	○7	○7	○7
○8	○8	○8	○8	○8	○8	○8	●8	○8
○9	○9	○9	○9	○9	○9	○9	○9	○9

Name of the Student: Irfan Hussain

Roll Number of the Student: 230123081

Signature of the Student: Irfan Hussain

Signature of the Invigilator: [illegible]

QUESTIONS AND RESPONSES

Question 1: (2 points) Consider the linear model

$$Y_1 = \beta_1 + \beta_2 + \epsilon_1$$
$$Y_2 = \beta_1 + 2\beta_2 + \epsilon_2$$
$$Y_3 = \beta_1 + c\beta_2 + \epsilon_3,$$

where $\beta_1, \beta_2 \in \mathbb{R}$ are unknown parameters, and the uncorrelated errors $\epsilon_i, i = 1,2,3$ have zero mean and finite variance $\sigma^2 (> 0)$. The constant c is such that $\hat{\beta}_1$ and $\hat{\beta}_2$ are uncorrelated, where $\hat{\beta}_1$ and $\hat{\beta}_2$ are the best linear unbiased estimators of β_1 and β_2, respectively. Then $(\text{Var}(\hat{\beta}_1), \text{Var}(\hat{\beta}_2))$ equals

0/2

- [x] $(\frac{\sigma^2}{3}, \frac{\sigma^2}{14})$ ✓
- [ ] $(3\sigma^2, \frac{\sigma^2}{14})$
- [x] $(\frac{\sigma^2}{14}, \frac{\sigma^2}{3})$ ✗
- [ ] $(\frac{\sigma^2}{3}, 14\sigma^2)$

Question 2: (3 points) Let $y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \cdots + \beta_{22} x_{22,i} + \epsilon_i, i = 1, 2, \cdots, 123$, where, $x_{1,i}, \cdots, x_{22,i}$ are fixed input (independent) variables; and, for $j = 0, 1, 2, \cdots, 22$; β_j are unknown parameters, and ϵ_i's are independent and identically distributed normal random variables with mean zero and finite variance. If $SS_{Reg} = 338.92$ and $SS_T = 522.30$ and the value of R^2_{adj} (adjusted R^2) equals to t. Then which of the following statements is/are true?

3/3

- [ ] $0.43 < t < 0.54$
- [ ] $0.61 < t < 0.69$
- [x] $0.53 \le t < 0.63$
- [x] $0.51 < t < 0.59$

Question 3: (3 points) Consider the multiple linear regression problem

$$Y = X\beta + \epsilon, \text{ and } \hat{Y} = X\hat{\beta}$$

where $Y = (Y_1, \cdots, Y_n)'$ is a $n \times 1$ vector of response variable, X is a $n \times p$ design matrix with full rank (largest possible rank), β is a $p \times 1$ vector of parameters, and ϵ is a $n \times 1$ vector of errors; $n > p$. The first element of β is the intercept of the model. Then which of the following statements is/are true?

3/3

- [x] $\sum_{i=1}^{n} Var(\hat{Y}_i) < \sigma^2 n$
- [ ] $\sum_{i=1}^{n} Var(\hat{Y}_i) = \sigma^2 n$
- [x] $\sum_{i=1}^{n} Var(\hat{Y}_i) = \sigma^2 p$
- [ ] $\sum_{i=1}^{n} Var(\hat{Y}_i) < \sigma^2 p$

Question 4: (2 points) Consider a multiple linear regression problem. Which of the following statements is/are true?

0/2

- [x] A linear model is linear in the parameters, but not necessarily in the independent variables
- [x] PRESS is a measure of how well a regression model perform in predicting new observations
- [x] Q-Q plot is used to test for normality
- [ ] A leverage point is always influential point

Question 5: (3 points) For $Y \in \mathbb{R}^n, X \in \mathbb{R}^{n \times p}$, and $\beta \in \mathbb{R}^p$, consider a regression model

$$Y = X\beta + \epsilon,$$

where ϵ has an n-dimensional multivariate normal distribution with zero mean vector and identity covariance matrix. Let I_p denote the identity matrix of order p. For $\lambda > 0$, let

$$\widehat{\beta}_n = (X^T X + \lambda I_p)^{-1} X^T Y,$$

be an estimator of β. Then which of the following options is/are correct?

3/3

- [ ] $\widehat{\beta}_n$ is an unbiased estimator of β
- [x] $(X^T X + \lambda I_p)$ is a positive definite matrix
- [x] $\widehat{\beta}_n$ has a multivariate normal distribution
- [ ] $\text{Var}(\widehat{\beta}_n) = (X^T X + \lambda I_p)^{-1}$

Question 6: (2 points) Let a linear model $Y = \beta_0 + \beta_1 X + \epsilon$ be fitted to the following data, where ϵ is normally distributed with mean 0 and unknown variance $\sigma^2 > 0$.

x_i	0	1	2	3	4
y_i	3	4	5	6	7

Let $\widehat{Y}_0$ denote the ordinary least-square estimator of Y at $X = 6$. If $E\left(\widehat{Y}\right) = c_1\beta_0 + c_2\beta_1$ and $Var\left(\widehat{Y}_0\right) = d\sigma^2$, then which of the following is/are true?

0/2

- [x] $c_1 + c_2 + d \leq 8.8$
- [x] $3.1 \leq c_2 - c_1 - d \leq 3.5$
- [x] $c_1 = 1,\ c_2 = 6,\ 1.7 \leq d \leq 1.9$
- [ ] $c_1 = 6,\ c_2 = 1,\ 1.7 \leq d \leq 1.9$